Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2023-24
FINITE ELEMENT METHODS

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

Q no.	Question	Marks	CO
a.	Define FEM?	2	1
b.	What is finite element analysis?	2	1
c.	What is Hermite shape function	2	1
d.	Write the limitation of FEM	2	1
e.	What is meant by axi-symmetric problems?	2	1
f.	What is the Global matrix?	2	1
g.	Write the stiffness matrix for the beam element	2	1
h.	Derive the shape functions for 2D truss element?	2	1
i.	Write the stiffness matrix for truss	2	1
j.	Describe the FEM formulation for 1D bar element	2	1

SECTION B

2. Attempt any *three* of the following:

Q no.	Question	Marks	CO
a.	Derive the shape function for linear rectangular element in local coordinate system; give its dimension of side ‘r’ and ‘s’ respectively	10	2
b.	Discuss the errors involved in finite element solution with the example of a model second order differential equation in one dimension.	10	2
c.	For a square, isotropic elastic body of thickness ‘h’, the displacement are approximated by: u (x, y) = (1 – x) yu1 + x(1 – y)u2 □(x, y) = 0	10	3
d.	Define the FEM software package like ANSYS, ABAQUS, NISA?	10	5
e.	What do you mean by isoperimetric formulation of a finite element problem? Give an example of real field problem where super parametric elements can be used and why?	10	3

SECTION C

3. Attempt any *one* part of the following:

a.	Discuss in detail about the concepts of FEM formulation How is that emerged as powerful tool. Discuss in detail about applications of finite element method	10	2
b.	Briefly discuss elimination approach to handle boundary conditions for solution of system of equations?	10	2

4. Attempt any *one* part of the following:

a.	Explain the Galerkin’s residual method and its use to derive the one-dimensional bar element equations.	10	3
b.	Using Rayleigh-Ritz method, find the expression for stress for one-dimensional element of length L subjected to an axial force q/unit length. AE is same throughout.	10	3

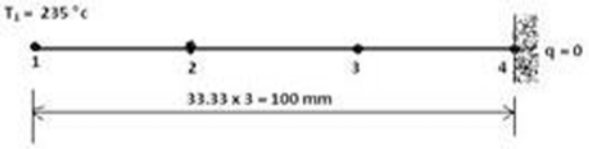
5. Attempt any *one* part of the following:

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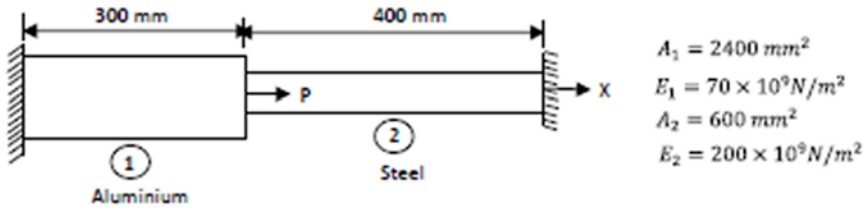
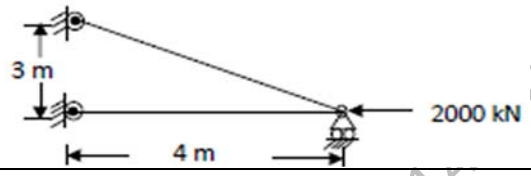
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a.	Derive Linear interpolation polynomials in terms of global coordinates for triangular (2D simplex) elements?	10	3
b.	A metallic fin with thermal conductivity K=360W/m^{0}c, 1mm thick and100mm long extends from a plane wall whose temperature is 235^{0}c. Determine the distribution and amount of heat transferred from the fin to air at 20^{0}c with h= 9W/m2^{0}c take width of the fin is 1000 mm. Assume tip is insulted 	10	4

6. Attempt any *one* part of the following:

a.	Determine the nodal displacements stress in each material and the reaction forces. For the bar shown in figure. An axial load P = 150 x10^3 N is applied as shown. 	10	4
b.	Determine the nodal displacement, element stresses and support reactions for the two-bar truss shown in figure. Take E = 210 GPa and A = 600 mm^2 for each element. 	10	4

7. Attempt any *one* part of the following:

a.	Formulate the finite element equations for Constant strain triangle shown in fig. Assume plane stress E=200 Gpa, ν=0.25, thickness=5mm, nodal co-ordinates. Pressure on 1-2 edge is 5N/mm^2 <table><tr><td>X1=1</td><td>X2=5</td><td>X3=3</td></tr><tr><td>Y1=2</td><td>Y2=4</td><td>Y3=6</td></tr></table>	10	4
b.	Derive the stiffness matrix for the four noded quadrilateral element in terms of natural coordinate systems	10	4

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